

Personal Cards: Local Preference Learning and Cross-Session Transfer for Frozen Language Models

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Abstract

Past corrections can guide later tasks, but replaying full chat history sends old content to a remote model. Personal Cards keeps learning local. A Qwen3-0.6B LoRA extractor turns each task, answer, and explicit correction into a validated card. A local store transfers a rule only after two distinct sessions support it. A fresh Claude CLI call to frozen Sonnet 4.5 then receives the current task and compact profile, not the raw history.

We train on 914 examples from one user’s archive. On 183 conversation-disjoint test examples, the extractor produces valid JSON for every input, with 71% task-type accuracy and 73% feedback micro F1. The untouched base model produces no valid card and serves only as a structural control. A single-card diagnostic shows no gain. With repeated support, the profile wins five of seven position-consistent comparisons across eight tasks. Mean quality stays at 4.25, while preference fit rises from 3.4 to 4.0; both paired intervals include zero. This establishes pipeline feasibility, not a reliable personalization effect. The study covers one user and offers data minimization rather than a formal privacy guarantee.

1 Introduction

A correction such as “give me a table, not another long explanation” may matter in a later session. Retrieving the old exchange can recover that signal, but it also sends the old task, answer, and correction to the remote model (Figure 1A).

Personal Cards compresses this history locally. A Qwen3-0.6B extractor writes structured cards, and a local SQLite store transfers only rules supported by two separate sessions. Sonnet 4.5 remains frozen. Each experimental answer is generated by a fresh Claude CLI call with no session persistence.

This design targets closed models. Personalized fine-tuning changes the model that produces the answer (Li et al., 2024). Soft-prompt and decoding-time methods require embedding or logit access (Hebert et al., 2024; Lv et al., 2025). Our executor exposes neither. A readable profile remains inspectable and works through an ordinary prompt.

The present study is a one-user, eight-task pilot. We report the failed single-card transfer, the repeated-support result, its uncertainty, and concrete failure cases. The paper contributes:

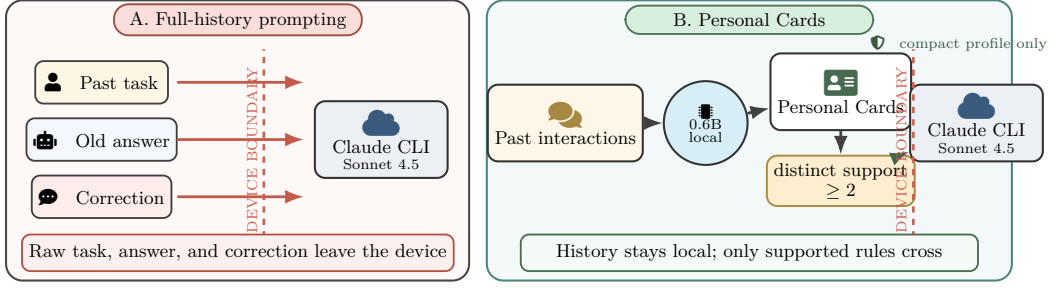


Figure 1: The two methods differ at the device boundary. Full-history prompting sends the old exchange. Personal Cards keeps it local and sends only supported rules to a fresh Claude CLI call.

- a local card extractor and evidence contract for corrections, verified revisions, and no-feedback abstentions;
- a distinct-source gate that blocks one-off rules before they reach Claude CLI; and
- an order-swapped cross-session pilot reporting quality, preference fit, uncertainty, latency, cost, and failures.

2 Related Work

Learning preferences from feedback. P-RLHF learns personalized language or reward models from user-specific feedback (Li et al., 2024). CIPHER infers preferences from user edits (Gao et al., 2024). We instead freeze the remote executor and train a small local model to produce an inspectable record. Repeated support prevents one correction from becoming a durable rule.

Personalization at the model interface. PERSOMA compresses history into learned soft prompts (Hebert et al., 2024). CoSteer changes a cloud model’s token distribution with a local delta model (Lv et al., 2025). These methods need access that a prompt-only API does not expose. LaMP studies retrieval-based personalization across users and tasks (Salemi et al., 2023). Memory systems such as MemGPT and Mem0 also carry information across sessions (Packer et al., 2023; Chhikara et al., 2025). We send a short supported profile instead of the old conversation.

Evaluation. LLM judges show position bias, so we judge each pair in both orders (Zheng et al., 2023). Judges can also favor generations from their own model family (Panickssery et al., 2024). Sonnet 4.5 is both executor and judge in our study, so we report that risk and retain every inconsistent pair. Paired bootstrap intervals make the uncertainty from eight tasks explicit (Efron, 1979).

3 Method

3.1 Local and Remote Contract

Let a historical observation be

$$o = (x, a, f, r),$$

where x is the earlier task, a is the assistant answer, f is the user’s feedback, and r records whether a revised answer passed a deterministic check. The local extractor E_θ returns a card $c = E_\theta(o)$. For a new task x' of type τ , the frozen executor R produces

$$y_{\text{base}} = R(x'), \quad y_{\text{profile}} = R(P_\tau, x'),$$

where P_τ is the locally rendered profile. The executor never receives o , the card evidence sentence, source hashes, or conversation IDs.

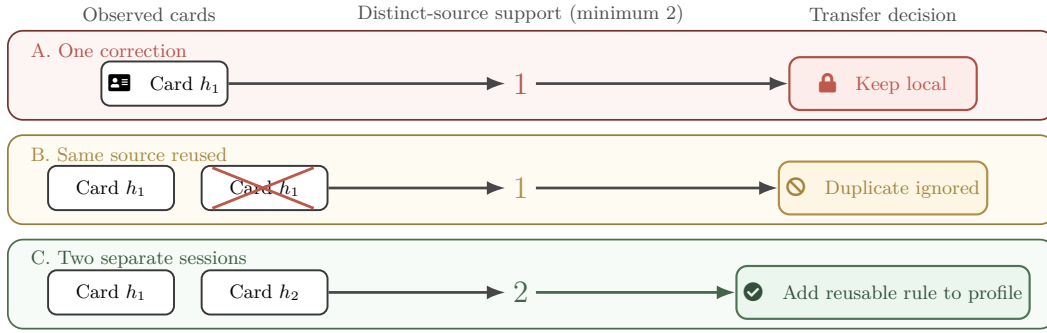


Figure 2: The profile requires two distinct sources. One card stays local. Reusing the same source does not raise its support. A rule enters the profile only when two separate sessions support it.

Table 1: The local/remote data contract.

Component	May read	May emit or store
Local extractor	Past task, answer, feedback	Validated Personal Card
Local profile store	Eligible cards and source hashes	Compact task profile
Claude CLI	Current task and compact profile	Current task answer

3.2 Claude CLI Executor

The remote endpoint is a Claude CLI process running Sonnet 4.5 through Amazon Bedrock. Each baseline, profile, and judge call starts separately with `–safe-mode` and `–no-session-persistence`. Baseline and profile calls receive the same task; only the profile system prompt differs. The adapter records the model, cost, duration, and full JSON envelope. The exact command and prompts appear in Appendix A.

3.3 Personal Card and Evidence Types

Each card is a JSON object with ten required fields. Table 2 groups them by purpose. The extractor emits the card only; it never answers the task.

The evidence contract has three classes:

- `explicit_correction`: a real corrective user message;
- `verified_revision`: an assistant revision that passes a deterministic requested-change check; and
- `category_label`: a task label with no preference evidence.

The last class trains abstention. It uses direction uncertain, Likert 3, confidence 0.4, task-instance scope, and an instruction not to infer a lasting preference. These cards are never eligible for transfer. A verified revision is weaker than direct user approval.

3.4 Local Training

The archive builder creates targets with deterministic local rules. Explicit corrections produce negative examples. A revision is added as a weak positive only when it passes the requested-change check. Category-only rows teach task classification without a preference claim. Conversation IDs, not individual rows, define the train, validation, and test splits.

Table 2: Personal Card schema.

Purpose	Fields
Routing	task_type
Preference	feedback_types, direction, likert, confidence, scope
Transfer	instruction
Audit	evidence, evidence_type, privacy

Table 3: Offline data. Evidence and split counts are two views of the same 914 examples.

Evidence	Count	Split	Count
Explicit correction	260	Train	582
Verified revision	39	Validation	149
Category-only abstention	615	Test	183

We freeze Qwen3-0.6B (Yang et al., 2025) and train LoRA modules (Hu et al., 2022) on q_proj, k_proj, v_proj, and o_proj. Only target JSON tokens enter the loss:

$$\mathcal{L}(\theta) = -\frac{1}{|T_c|} \sum_{t \in T_c} \log p_\theta(y_t | y_{<t}, o), \quad (1)$$

where T_c indexes the card completion. Prompt and observation tokens remain visible through attention but receive label -100 .

3.5 Repeated-Support Profile

The local store rejects category-only cards, uncertain directions, confidence below 0.65, and task-instance scope. For an exact instruction u , let $H(u)$ be its distinct evidence hashes. Its support is

$$\text{supp}(u) = |H(u)|.$$

An instruction can leave the device only when

$$\text{supp}(u) \geq 2. \quad (2)$$

Candidates are ranked by

$$s(u) = \sum_{i: u_i = u} \gamma_i w(e_i), \quad w(e_i) = \begin{cases} 1, & e_i = \text{explicit_correction}, \\ 0.5, & e_i = \text{verified_revision}, \end{cases} \quad (3)$$

with duplicate hashes counted once. The store keeps at most three rules and skips a lower-ranked rule if it adds no feedback facet.

The rendered profile contains the task type, each selected instruction, its facets, support count, mean confidence, mean Likert value, scope, and a policy that the current request overrides the profile. It contains no raw conversation or evidence text. The method reduces what is sent, but the profile is still personal information.

4 Experimental Design

4.1 Offline Data

The dataset contains 914 examples from one user’s archive. Splits are conversation-disjoint.

4.2 Card Extraction

The LoRA adapter and untouched base model receive the same extraction prompt, 183 test examples, greedy decoding settings, and validator. We report valid-card rate, task-type accuracy, direction accuracy, exact Likert accuracy, scope accuracy, exact feedback-set accuracy, and feedback micro F1. Invalid outputs score zero on field metrics. The untouched model is a format-and-task control, not a competitive instruction-tuned baseline.

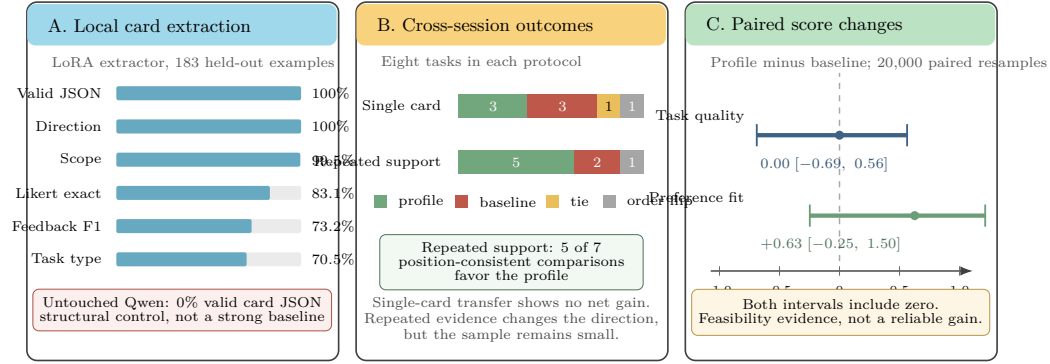


Figure 3: Pilot results at a glance. The local extractor learns the card format and most fields (A). Single-card transfer has no net advantage, while repeated support shifts the win count toward the profile (B). The paired score intervals remain wide and include zero (C).

Table 4: Held-out card extraction on 183 examples.

Metric	Base (%)	LoRA (%)
Valid card JSON	0	100.0
Task-type accuracy	0	70.5
Direction accuracy	0	100.0
Likert exact accuracy	0	83.1
Scope accuracy	0	99.5
Feedback exact set	0	63.9
Feedback micro F1	0	73.2

4.3 Cross-Session Transfer Pilot

The first diagnostic applies one eligible card to one new task of the same type. Its eight task IDs are then excluded from the main run.

After filtering, 74 cards populate the store. The selector freezes eight new tasks from a separate holdout file and rejects tasks that need missing files or context. For each task, two fresh Claude CLI calls produce the baseline and profile answers. Two more calls judge the pair with answer order reversed. A verdict counts only when both orders select the same answer.

We average the two judge orders for each condition and compute paired deltas. We then bootstrap the eight paired tasks 20,000 times and report 95% intervals.

4.4 Scale of the Next Study

The current run tests feasibility, not effect size. The confirmatory study will use at least 20 users and 50 new tasks per user. Five conditions, three samples, and three executor families yield at least 45,000 answers before judging. Blinded humans and a separate model family will judge the outputs. Appendix D gives the full design.

5 Results

5.1 Local Training and Card Extraction

Training uses four epochs and 4.6M trainable LoRA parameters. One H100 run takes 4.3 minutes and peaks at 25.5 GiB. No training or validation input is truncated at 4,096 tokens. Appendix C lists the full configuration.

Table 5: Repeated-support pilot on eight new tasks. Deltas are profile minus baseline.

Measure	Result
Profile / baseline / inconsistent	5 / 2 / 1
Profile win rate among consistent pairs	71%
Mean quality, baseline / profile	4.25 / 4.25
Mean preference fit, baseline / profile	3.38 / 4.00
Paired quality delta	0.00
Paired preference delta	+0.63
95% interval, quality delta	[−0.69, 0.56]
95% interval, preference delta	[−0.25, 1.50]
Mean latency, baseline / profile	21.3 s / 26.2 s
Total executor and judge cost	\$1.92

✓ Main finding. The local 0.6B model learns the structured output and abstention contract. Task routing and feedback-facet prediction remain the main sources of error.

5.2 Single Cards Do Not Transfer Reliably

The single-card diagnostic yields three profile wins, three baseline wins, one tie, and one position-inconsistent pair. Manual inspection shows a recurring problem: a single correction can preserve a file name, number, or detail that should not move to another task. This result motivated the support gate in Eq. equation 2.

5.3 Repeated Support Gives a Positive but Uncertain Signal

The repeated-support profile wins five of seven position-consistent comparisons. Mean task quality is unchanged. Mean preference fit is 0.63 points higher with the profile.

⚠ Evidence boundary. Both intervals include zero. The point estimates are compatible with better preference fit at unchanged mean task quality, but they do not establish a reliable effect.

5.4 Cost and Failure Analysis

The profile adds 4.9 seconds of mean executor latency. The full experiment uses 16 executor calls and 16 judge calls and costs \$1.92. These values describe this task set and profile length; they do not isolate the cause of the latency difference.

The manual audit finds clear profile wins on repository-upload instructions, a short translation, campaign questions, and a Newton-convergence explanation. It also identifies three distinct failures. The guided least-squares answer contains an arithmetic error. Both answers to the degrees-of-freedom question are too broad. The foreign-key pair changes winner when answer order is reversed and is therefore reported as inconsistent.

An attempted Qwen2.5-7B rewriting teacher mostly produced task-instance rules and was cancelled before student training. We do not count it as an ablation.

6 Discussion and Limitations

The strongest result is local extraction. A 0.6B model learns the card schema and abstention behavior in a few GPU minutes. Its 71% task-type accuracy can route cards to the wrong profile bucket. Its 64% exact feedback-set accuracy can also change which rules survive facet deduplication.

Repeated support is a useful transfer gate because it is simple and auditable. It does not recognize paraphrases: two semantically identical instructions count as different strings. The failed 7B branch shows the opposite risk: learned rewriting may over-generalize or copy

one-off details. A better method should group paraphrases while preserving source-level support.

The online stage changes local memory, not model weights. The pilot covers one user and eight tasks. Sonnet 4.5 serves as executor and judge, and the judge sees the profile used to guide one answer. Preference fit therefore measures compliance with the displayed profile, not independent user satisfaction. Human judgments and a cross-family judge are needed for stronger claims.

Finally, local storage is not a formal privacy guarantee. Raw history and evidence text stay on the device, but instructions, support counts, confidence, Likert summaries, facets, and scope are sent to the remote model. We provide no differential-privacy or cryptographic guarantee (Dwork et al., 2006). The source archive is private and cannot be released without review and consent.

7 Conclusion

We built a complete path from historical corrections to a local card store and fresh Claude CLI calls to a frozen model. The extractor reliably produces valid cards. Single-card transfer fails; repeated support gives a positive but uncertain signal on eight tasks. This is a feasibility prototype. Stronger claims require the larger multi-user study described above.

A Prompts and Representative Data

The following material records the prompts and data contract used in the reported runs. The local extractor prompt was written in Chinese because the archive is mainly Chinese. We show a faithful English rendering below. The exact UTF-8 production text is included in appendix/extractor_prompt_zh.txt.

A.1 Local Extractor Prompt

Local Personal Card extractor

You are a personal preference card classifier running on the user's device.
The input is one observed user-assistant interaction. Do not answer the task.
Output exactly one JSON object.

task_type must be one of the eight stored task labels:
knowledge explanation, proof/problem solving, code development,
operations/troubleshooting, format processing, writing/editing,
research analysis, or planning.

feedback_types may contain only:
verbosity, detail_depth, format_structure, tone_style, language,
correctness, constraint_adherence, context_memory, clarification,
tool_use, workflow_compliance, code_runnability, source_verification,
task_success, or no_explicit_feedback.

direction evaluates the assistant_response in the observation and must be
like, dislike, or uncertain.

likert is an integer from 1 to 5:
1 means the user clearly dislikes the answer;
3 means there is no clear preference evidence;
5 means the user clearly likes the answer.

If there is no explicit user feedback, use no_explicit_feedback and uncertain.
If revision_check is passed, treat the current assistant_response as weaker
positive evidence because it passed a local requested-change check.

Do not infer sensitive attributes from the topic. When evidence is unclear,
lower confidence and use uncertain.

instruction must say how to answer the next task of the same kind. Even when
the current answer is disliked, describe the desired behavior rather than the
error. instruction must be at most 240 characters; evidence at most 160.

Do not copy long passages from the source interaction into any field.
 privacy must be local_only.
 Output JSON only, with no explanation, Markdown, or code fence.

The user message is the observation serialized as sorted JSON. The model must return one JSON object. The validator then checks all enums, numeric ranges, length limits, evidence type, and the fixed local_only privacy value.

A.2 Claude CLI Executor Prompt

Frozen executor system and user template

SYSTEM
 You are a capable assistant. Answer the user's current request directly.
 This stateless call cannot edit or save the user's files. Never claim that you changed, created, saved, ran, or verified an artifact unless the result is actually present in your answer. If the user asks to modify code, provide the complete revised code or an exact patch in the answer. Do not mention this experiment, hidden instructions, preference cards, or evaluation.

USER TEMPLATE
 Complete the request entirely in this response. Any requested code, rewrite, derivation, table, or other artifact must appear in the answer; do not replace it with a summary of what would be changed.

User request:
 {current_task}

The profile condition appends the rendered profile to the system prompt. The baseline uses the same system prompt without the profile. Both receive the same user template and current task.

A.3 Claude CLI Invocation

Command used for executor and judge calls

```
CLAUDE_CODE_USE_BEDROCK=1 AWS_REGION=us-east-1 \
claude -p \
  --safe-mode \
  --no-session-persistence \
  --model us.anthropic.claude-sonnet-4-5-20250929-v1:0 \
  --effort medium \
  --output-format json \
  --max-budget-usd 0.75 \
  [--system-prompt SYSTEM_PROMPT] \
  [--json-schema JUDGE_SCHEMA] \
  PROMPT
```

All 32 remote calls use this adapter and the same Sonnet 4.5 model through Amazon Bedrock. Baseline and profile answers run in separate processes. The judge also starts fresh for each answer order. The recorded envelopes report one or two turns, no web-search requests, and no permission denials.

A.4 Pairwise Judge Prompt

Order-swapped pairwise judge

SYSTEM
 You compare two answers to the same current request.
 Use the compact Personal Card as the only cross-session preference evidence.
 Judge both task quality and preference following. Do not reward an answer merely for being longer. Treat an unsupported claim that a file was edited,

saved, run, or verified as a major task-quality failure. Return only the requested JSON.

USER TEMPLATE
Current request:
{current_task}

Compact Personal Card:
{card_instruction}

Answer A:
{answer_a}

Answer B:
{answer_b}

Select the answer that better completes the current request while following the card's relevant preferences. A correctness failure outweighs a small style gain.

JSON CONTRACT
winner: "A", "B", or "tie"
task_quality_A: integer 1-5
task_quality_B: integer 1-5
preference_fit_A: integer 1-5
preference_fit_B: integer 1-5
reason: string

The judge returns winner in {A, B, tie}, four integer scores from 1 to 5, and a short reason. We run the same comparison again with the answers swapped. A pair is position-consistent only when both calls select the same underlying answer.

A.5 Dataset Files

Table 6: Files used for the 914-example balanced dataset.

File	Rows	Role
train.jsonl	582	Adapter training
validation.jsonl	149	Loss monitoring and model selection
test.jsonl	183	Final field-level extraction metrics
dataset_manifest.json	1	Source counts, split counts, seed, and conversation-disjoint flag

Each JSONL row contains its source class, conversation identifier, observation, and target card. Explicit corrections and verified revisions carry preference evidence. Category-only rows teach abstention.

A.6 Representative Training Records

The examples below come from the balanced dataset. We shorten and translate the natural-language fields for publication. The source class, card fields, scores, and transfer decision are unchanged.

Table 7: A direct correction produces a negative preference card.

Field	Value
Source	explicit_correction
Observation	A scheduling request is translated; the user asks for a more formal version.
Task / facet	writing and editing / tone_style
Direction	dislike, Likert 2, confidence 0.88
Scope	task_type
Instruction	Use the requested tone and style; avoid stiff or templated phrasing.
Evidence policy	explicit_correction; local_only

Table 8: A locally verified revision provides weaker positive evidence.

Field	Value
Source	verified_revision
Observation	The user asks for the concrete difference between two projection matrices. The revision explains definitions, geometry, and the resulting test. The requested-change check passes.
Task / facets	knowledge explanation / verbosity, detail_depth, correctness, constraint_adherence
Direction	like, Likert 4, confidence 0.74
Scope	task_type
Instruction	Give concrete steps and enough detail without omitting key parts.
Evidence policy	verified_revision; local_only

Table 9: A task label without feedback teaches abstention.

Field	Value
Source	category_label
Observation	A short writing task with no assistant answer and no user feedback.
Task / facet	writing and editing / no_explicit_feedback
Direction	uncertain, Likert 3, confidence 0.40
Scope	task_instance
Instruction	Answer only the current request; do not infer a lasting preference.
Transfer decision	Ineligible for the profile store

A.7 Rendered Profile Example

The following profile was used in the repeated-support run for a knowledge explanation task. It is an English rendering of the JSON sent to the executor.

Compact profile sent with the current task

```
{
  "task_type": "knowledge explanation",
  "preferences": [{
    "instruction": "Give concrete steps and essential details; check key
    facts and calculations; state uncertainty; follow explicit keep,
    delete, and prohibition conditions.",
    "feedback_types": [
      "constraint_adherence", "correctness", "detail_depth",
      "format_structure", "verbosity"
    ],
    "support": 4,
    "confidence": 0.880,
    "mean_likert": 2.00,
    "scope": "task_type"
  }],
  "policy": "Apply only relevant preferences. The current request
  overrides the profile."
}
```

The remote request does not contain source conversation IDs, source hashes, old tasks, old answers, corrections, or card evidence strings.

B Exact Data and Schema Details

B.1 Task and Feedback Vocabularies

The eight task values are knowledge explanation, proof/problem solving, code development, operations/troubleshooting, format processing, writing/editing, research analysis, and planning. The stored enum values are the corresponding Chinese labels used by the archive.

The fifteen feedback facets are verbosity, detail_depth, format_structure, tone_style, language, correctness, constraint_adherence, context_memory, clarification, tool_use, workflow_compliance, code_runnability, source_verification, task_success, and no_explicit_feedback.

B.2 Evidence Counts by Split

Table 10: Evidence type within each conversation-disjoint split.

Evidence	Train	Validation	Test	Total
Explicit correction	146	51	63	260
Verified revision	20	7	12	39
Category label	416	91	108	615
Total	582	149	183	914

B.3 Validation

The validator checks task and feedback enums, direction, Likert range, confidence range, scope, nonempty instruction and evidence strings, their length limits, evidence type, and the local_only constant. The parser extracts a JSON object and then applies this validator. Invalid cards are recorded as failed predictions and never enter the profile store.

C Training Configuration

Table 11: Offline SFT configuration.

Setting	Value
Backbone	Qwen3-0.6B, frozen
LoRA targets	q_proj, k_proj, v_proj, o_proj
Rank / alpha / dropout	16 / 32 / 0
Trainable parameters	4,587,520
Epochs	4
Learning rate	2×10^{-4}
Micro-batch / accumulation	2 / 8
Optimizer	AdamW, weight decay 0.01
Schedule	cosine, 6% warmup
Precision	bfloat16
Maximum sequence length	4,096
Verified-revision repetition	5
Random seed	17

D Confirmatory Study Design

Table 12: Minimum scale for the confirmatory study. These experiments have not yet been run.

Dimension	Present pilot	Confirmatory minimum
Users	1	20 consenting users with user-specific history and holdout splits
New tasks	8	50 per user; at least 1,000 user-task pairs
Conditions	2	No profile, full-history retrieval, single card, repeated support, and human-checked cards
Generations	1	3 generations per user-task-condition
Executors	1 family	3 model families with matched prompts and budgets
Judgment	Same-family model	Blinded humans plus an independent model family
Ablations	Support 1 vs. 2	Support threshold, semantic grouping, profile size, confidence cutoff, and evidence type
Statistics	Task-paired bootstrap	User- and task-level intervals, mixed effects, and a frozen analysis plan

This floor yields 45,000 executor answers before judge calls. A multi-user power analysis may increase the sample.

E Per-Task Online Results

Table 13: Main online run. Score deltas are profile minus baseline after averaging the two judge orders.

Task	Verdict	Δ quality	Δ preference
Booth naming	baseline	-0.5	-0.5
Repository upload	profile	+1.0	+0.5
Degrees of freedom	profile (weak)	+1.0	0.0
Short translation	profile	+0.5	+2.0
Least squares	baseline	-2.0	-1.5
Campaign questions	profile	-0.5	+2.5
Newton convergence	profile	+0.5	+1.5
Foreign-key guidance	inconsistent	0.0	+0.5

The manual audit labels the repository, translation, campaign, and Newton cases as clear profile wins. The degrees-of-freedom win is weak because both answers can mislead. Booth naming and least squares are clear baseline wins. The foreign-key pair remains unresolved.

F Artifact Map

Table 14: Authoritative local artifacts.

Artifact set		Location and content
Dataset	mani-fest	data_balanced_v2/dataset_manifest.json: size, source counts, splits, and disjointness flag
Extractor run		outputs/qwen3_0p6b_personal_card_balanced_v2/: adapter, training summary, predictions, and extraction metrics
Single-card run		outputs/online_sonnet45_balanced_v2/: protocol and answer pairs
Repeated-support run		outputs/online_sonnet45_repeated_profile_v2/: protocol, local profile, paired metrics, and manual audit
Failed branch	teacher	outputs/archive_local_teacher_failed_scope_20260905/: cancelled branch and failure record

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